

EX 1 Calculer et donner un résultat simplifié.

1. $A = \frac{-4}{-5} \times \frac{6}{7} + \frac{-1}{7}$

2. $B = \frac{-2}{7} - \frac{2}{5} \times \frac{1}{7}$

3. $C = \frac{-6}{7} \div \frac{-4}{7} + \frac{-1}{5}$

4. $D = \frac{-4}{-7} - \frac{1}{5} \div \frac{2}{3}$

5. $E = \frac{1}{7} \times \frac{1}{7} - \frac{2}{7}$

6. $F = \frac{1}{-5} - \frac{4}{-5} \times \frac{-1}{-7}$

EX 2 Calculer et donner un résultat simplifié.

1. $A = \frac{2}{-3} + \frac{-5}{-3} \div \frac{-1}{8}$

2. $B = \frac{-5}{6} \div \frac{-7}{-5} - \frac{-4}{-5}$

3. $C = \frac{2}{9} - \frac{3}{9} \div \frac{5}{-7}$

4. $D = \frac{2}{7} \div \frac{4}{-9} - \frac{-4}{-9}$

5. $E = \frac{8}{-9} + \frac{-3}{-9} \times \frac{6}{-7}$

6. $F = \frac{5}{9} \div \frac{1}{-9} + \frac{2}{-9}$

EX 3 Calculer et donner un résultat simplifié.

1. $A = \frac{-1}{9} + \frac{-1}{9} \times \frac{-3}{4}$

2. $B = \frac{2}{-5} + \frac{-3}{-5} \times \frac{-3}{4}$

3. $C = \frac{-2}{-9} + \frac{4}{-9} \div \frac{-5}{-8}$

4. $D = \frac{3}{7} - \frac{-5}{7} \div \frac{-7}{10}$

5. $E = \frac{1}{-4} + \frac{3}{-4} \div \frac{5}{6}$

6. $F = \frac{-1}{-5} - \frac{5}{-5} \times \frac{5}{8}$



EX

1

$$\begin{aligned}
 1. \quad A &= \frac{-4}{-5} \times \frac{6}{7} + \frac{-1}{7} \\
 A &= \frac{-4}{-5} \times \frac{6}{7} + \frac{-1}{7} \\
 A &= \frac{-4 \times 6}{-5 \times 7} + \frac{-1}{7} \\
 A &= \frac{-24}{-35} + \frac{-1}{7} \\
 A &= \frac{-24}{-35} + \frac{-1 \times (-5)}{7 \times (-5)} \\
 A &= \frac{-24}{-35} + \frac{5}{-35} \\
 A &= \frac{-19}{-35} \\
 A &= \frac{19}{35}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad B &= \frac{-2}{7} - \frac{2}{5} \times \frac{1}{7} \\
 B &= \frac{-2}{7} - \frac{2 \times 1}{5 \times 7} \\
 B &= \frac{-2}{7} - \frac{2}{35} \\
 B &= \frac{-2 \times 5}{7 \times 5} - \frac{2}{35} \\
 B &= \frac{-10}{35} - \frac{2}{35} \\
 B &= \frac{-12}{35} \\
 B &= -\frac{12}{35}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad C &= \frac{-6}{7} \div \frac{-4}{7} + \frac{-1}{5} \\
 C &= \frac{-6}{7} \times \frac{7}{-4} + \frac{-1}{5} \\
 C &= \frac{-6}{7} \times \frac{7}{-4} + \frac{-1}{5} \\
 C &= \frac{-6 \times 7}{7 \times (-4)} + \frac{-1}{5} \\
 C &= \frac{-42}{-28} + \frac{-1}{5} \\
 C &= \frac{-42 \times 5}{-28 \times 5} + \frac{-1 \times (-28)}{5 \times (-28)} \\
 C &= \frac{-210}{-140} + \frac{28}{-140} \\
 C &= \frac{-140}{-140} \\
 C &= \frac{13 \times 14}{10 \times 14} \\
 C &= \frac{13}{10}
 \end{aligned}$$

$$\begin{aligned}
 4. \quad D &= \frac{-4}{-7} - \frac{1}{5} \div \frac{2}{3} \\
 D &= \frac{-4}{-7} - \frac{1}{5} \times \frac{3}{2} \\
 D &= \frac{-4}{-7} - \frac{1 \times 3}{5 \times 2} \\
 D &= \frac{-4}{-7} - \frac{3}{10} \\
 D &= \frac{-4 \times 10}{-7 \times 10} - \frac{3 \times (-7)}{10 \times (-7)} \\
 D &= \frac{-40}{-70} - \frac{-21}{-70} \\
 D &= \frac{-19}{-70} \\
 D &= \frac{19}{70}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad E &= \frac{1}{7} \times \frac{1}{7} - \frac{2}{7} \\
 E &= \frac{1}{7} \times \frac{1}{7} - \frac{2}{7} \\
 E &= \frac{1 \times 1}{7 \times 7} - \frac{2}{7} \\
 E &= \frac{1}{49} - \frac{2}{7} \\
 E &= \frac{1}{49} - \frac{2 \times 7}{7 \times 7} \\
 E &= \frac{1}{49} - \frac{14}{49} \\
 E &= \frac{-13}{49} \\
 E &= -\frac{13}{49}
 \end{aligned}$$

$$\begin{aligned}
 6. \quad F &= \frac{1}{-5} - \frac{4}{-5} \times \frac{-1}{7} \\
 F &= \frac{1}{-5} - \frac{4 \times (-1)}{-5 \times (-7)} \\
 F &= \frac{1}{-5} - \frac{-4}{35} \\
 F &= \frac{1 \times 7}{-5 \times 7} - \frac{-4 \times (-1)}{35 \times (-1)} \\
 F &= \frac{7}{-35} - \frac{4}{-35} \\
 F &= \frac{3}{-35} \\
 F &= -\frac{3}{35}
 \end{aligned}$$



EX

2

$$\begin{aligned}
 1. \quad A &= \frac{2}{-3} + \frac{-5}{-3} \div \frac{-1}{8} \\
 A &= \frac{2}{-3} + \frac{-5}{-3} \times \frac{8}{-1} \\
 A &= \frac{2}{-3} + \frac{-5 \times 8}{-3 \times (-1)} \\
 A &= \frac{2}{-3} + \frac{-40}{3} \\
 A &= \frac{2}{-3} + \frac{-40 \times (-1)}{3 \times (-1)} \\
 A &= \frac{2}{-3} + \frac{40}{-3} \\
 A &= \frac{42}{-3} \\
 A &= -14
 \end{aligned}$$

$$\begin{aligned}
 2. \quad B &= \frac{-5}{6} \div \frac{-7}{-5} - \frac{-4}{-5} \\
 B &= \frac{-5}{6} \times \frac{-5}{-7} - \frac{-4}{-5} \\
 B &= \frac{-5}{6} \times \frac{-5}{-7} - \frac{-4}{-5} \\
 B &= \frac{-5 \times (-5)}{6 \times (-7)} - \frac{-4}{-5} \\
 B &= \frac{25}{-42} - \frac{-4}{-5} \\
 B &= \frac{25 \times (-5)}{-42 \times (-5)} - \frac{-4 \times (-42)}{-5 \times (-42)} \\
 B &= \frac{-125}{210} - \frac{168}{210} \\
 B &= \frac{-293}{210} \\
 B &= -\frac{293}{210}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad C &= \frac{2}{9} - \frac{3}{9} \div \frac{5}{-7} \\
 C &= \frac{2}{9} - \frac{3}{9} \times \frac{5}{-7} \\
 C &= \frac{2}{9} - \frac{3 \times 5}{9 \times (-7)} \\
 C &= \frac{2}{9} - \frac{-21}{45} \\
 C &= \frac{2 \times 5}{9 \times 5} - \frac{-21}{45} \\
 C &= \frac{10}{45} - \frac{-21}{45} \\
 C &= \frac{31}{45}
 \end{aligned}$$

$$4. \quad D = \frac{2}{7} \div \frac{4}{-9} - \frac{-4}{-9}$$

$$\begin{aligned}
 D &= \frac{2}{7} \times \frac{-9}{4} - \frac{-4}{-9} \\
 D &= \frac{2}{7} \times \frac{-9}{4} - \frac{-4}{-9} \\
 D &= \frac{2 \times (-9)}{7 \times 4} - \frac{-4}{-9} \\
 D &= \frac{-18}{28} - \frac{-4}{-9} \\
 D &= \frac{-18 \times (-9)}{28 \times (-9)} - \frac{-4 \times 28}{-9 \times 28} \\
 D &= \frac{162}{-252} - \frac{-112}{-252} \\
 D &= \frac{274}{-252} \\
 D &= -\frac{137 \times 2}{126 \times 2} \\
 D &= -\frac{137}{126}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad E &= \frac{8}{-9} + \frac{-3}{-9} \times \frac{6}{-7} \\
 E &= \frac{8}{-9} + \frac{-3 \times 6}{-9 \times (-7)} \\
 E &= \frac{8}{-9} + \frac{-18}{63} \\
 E &= \frac{8 \times 7}{-9 \times 7} + \frac{-18 \times (-1)}{63 \times (-1)} \\
 E &= \frac{56}{-63} + \frac{18}{-63} \\
 E &= \frac{74}{-63} \\
 E &= -\frac{74}{63}
 \end{aligned}$$

$$\begin{aligned}
 6. \quad F &= \frac{5}{9} \div \frac{1}{-9} + \frac{2}{-9} \\
 F &= \frac{5}{9} \times -9 + \frac{2}{-9} \\
 F &= \frac{5}{9} \times -9 + \frac{2}{-9} \\
 F &= \frac{5 \times (-9)}{9 \times 1} + \frac{2}{-9} \\
 F &= \frac{-45}{9} + \frac{2}{-9} \\
 F &= \frac{-45 \times (-1)}{9 \times (-1)} + \frac{2}{-9} \\
 F &= \frac{45}{-9} + \frac{2}{-9} \\
 F &= \frac{47}{-9} \\
 F &= -\frac{47}{9}
 \end{aligned}$$

EX
3

$$\begin{aligned}
 1. \quad A &= \frac{-1}{9} + \frac{-1}{9} \times \frac{-3}{4} \\
 A &= \frac{-1}{9} + \frac{-1 \times (-3)}{9 \times 4} \\
 A &= \frac{-1}{9} + \frac{3}{36} \\
 A &= \frac{-1 \times 4}{9 \times 4} + \frac{3}{36} \\
 A &= \frac{-4}{36} + \frac{3}{36} \\
 A &= \frac{-1}{36} \\
 A &= -\frac{1}{36}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad B &= \frac{2}{-5} + \frac{-3}{-5} \times \frac{-3}{4} \\
 B &= \frac{2}{-5} + \frac{-3 \times (-3)}{-5 \times 4} \\
 B &= \frac{2}{-5} + \frac{9}{-20} \\
 B &= \frac{2 \times (-4)}{-5 \times (-4)} + \frac{9 \times (-1)}{-20 \times (-1)} \\
 B &= \frac{-8}{20} + \frac{-9}{20} \\
 B &= \frac{-17}{20} \\
 B &= -\frac{17}{20}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad C &= \frac{-2}{-9} + \frac{4}{-9} \div \frac{-5}{-8} \\
 C &= \frac{-2}{-9} + \frac{4}{-9} \times \frac{-8}{-5} \\
 C &= \frac{-2}{-9} + \frac{4 \times (-8)}{-9 \times (-5)} \\
 C &= \frac{-2}{-9} + \frac{-32}{45} \\
 C &= \frac{-2 \times 5}{-9 \times 5} + \frac{-32 \times (-1)}{45 \times (-1)} \\
 C &= \frac{-10}{-45} + \frac{32}{-45} \\
 C &= \frac{22}{-45} \\
 C &= -\frac{22}{45}
 \end{aligned}$$

$$\begin{aligned}
 4. \quad D &= \frac{3}{7} - \frac{-5}{7} \div \frac{-7}{10} \\
 D &= \frac{3}{7} - \frac{-5}{7} \times \frac{10}{-7} \\
 D &= \frac{3}{7} - \frac{-5 \times 10}{7 \times (-7)} \\
 D &= \frac{3}{7} - \frac{-50}{-49} \\
 D &= \frac{3 \times (-7)}{7 \times (-7)} - \frac{-50}{-49} \\
 D &= \frac{-21}{-49} - \frac{-50}{-49} \\
 D &= \frac{29}{-49} \\
 D &= -\frac{29}{49}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad E &= \frac{1}{-4} + \frac{3}{-4} \div \frac{5}{6} \\
 E &= \frac{1}{-4} + \frac{3}{-4} \times \frac{6}{5} \\
 E &= \frac{1}{-4} + \frac{3 \times 6}{-4 \times 5} \\
 E &= \frac{1}{-4} + \frac{18}{-20} \\
 E &= \frac{1 \times (-5)}{-4 \times (-5)} + \frac{18 \times (-1)}{-20 \times (-1)} \\
 E &= \frac{-5}{20} + \frac{-18}{20} \\
 E &= \frac{-23}{20} \\
 E &= -\frac{23}{20}
 \end{aligned}$$

$$\begin{aligned}
 6. \quad F &= \frac{-1}{-5} - \frac{5}{-5} \times \frac{5}{8} \\
 F &= \frac{-1}{-5} - \frac{5 \times 5}{-5 \times 8} \\
 F &= \frac{-1}{-5} - \frac{25}{-40} \\
 F &= \frac{-1 \times (-8)}{-5 \times (-8)} - \frac{25 \times (-1)}{-40 \times (-1)} \\
 F &= \frac{8}{40} - \frac{-25}{40} \\
 F &= \frac{33}{40}
 \end{aligned}$$